Finandex Whitepaper

Global Financial Market Indexing Insights — 27 data-grounded chatbots for markets, economics, and law (including financial law & help for scam victims)

Version 2.0 · 23 October 2026

by Dea Saka Kurnia Putra

© 2026 Dea Saka Kurnia Putra. All rights reserved.

Contents

1. Executive summary
2. Background and problem
3. Goals and design principles
4. Product and user experience
5. The 27 Dex
6. My LAIwyer: financial law and help for scam victims
7. Data and pipeline
8. Analytics methodology
9. System architecture
10. How the chatbot works
11. Tech stack
12. Quality and verification
13. Security and privacy
14. Deployment and operations
15. Data licensing and distribution
16. Limitations
17. Roadmap
18. Closing and disclaimer

© 2026 Dea Saka Kurnia Putra. All rights reserved.

1. Executive summary

Finandex is a progressive web app (PWA) with Android and iOS versions that offers 27 data-grounded chatbots — called "Dex" — for global financial markets, Indonesian markets and economy, crypto and macro, startups and venture capital, cost of living, social issues, public sentiment, and legal information, including financial-sector law and help for scam victims.

Every number comes from a verified dataset, is computed by code tested against reference (golden) values, and is always shown with a chart, its source, its as-of date, and how it was calculated. Every legal quote comes from the official text of a regulation, with document, article, and page.

Metric	Value
Dex	27 in 8 categories
Input datasets	42 files verified by SHA-256
Legal corpus	4 core documents + 19 official financial-sector texts
Analytics tools	362 read-only tools
Output data	≈ 680 MB (core ≈ 33 MB, available offline)
Languages	Indonesian & English

© 2026 Dea Saka Kurnia Putra. All rights reserved.

2. Background and problem

Market and economic data is scattered across many sources with different formats, quality, and dates. General chatbots answer fluently but often cannot show where a number came from, how current it is, or how it was computed. In law, answers from a model's memory risk citing the wrong or repealed article.

Financial transaction fraud (phishing/OTP, fraudulent investment, illegal online lending, money mules) keeps growing. Victims need the right immediate steps and official reporting channels, while the legal basis is spread over a dozen laws and regulations.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

3. Goals and design principles

- Numbers only from data — no invented numbers or URLs.
- Traceable — source file, as-of date, formula, and rows used are always visible.
- One computation path — chat and feature tabs use the same tools.
- Safe — no accounts, no tracking, sensitive-data blocking, prompt-injection defences.
- Inclusive — WCAG 2.2 AA, two languages, offline, phone and desktop.
- Honest about limits — not real-time, samples, synthetic data, and "not advice" stated openly.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

4. Product and user experience

Area	Contents
Home	Ask box, cross-category suggestions, 6-asset market summary with sparklines, whitepaper card, feature map of 27 Dex (search, categories, Indonesia filter, hide non-commercial Dex), "As of" strip.
Dex page	Header with as-of date, quality, status, licence, sample; Chat tab (first), feature tabs, and Data tab (last).
Chat	Streaming, tool status, answer card (core answer → chart → analysis → "Why this answer?" → limitations & citations), stop/retry/copy/share, history per Dex.
Charts & tables	Line, candlestick, bar, heatmap, network graph, map (bar/scatter), legal quote card; full screen, watermarked PNG/CSV export, "Ask about this".
Global search	Ctrl/Cmd+K: Dex, assets, stocks, companies, funds, coins, and "search in Dex".
Other	Cross-Dex watchlist, settings (theme, language, colour-blind, delete data), About, Legal, Whitepaper (PDF download ID/EN).

© 2026 Dea Saka Kurnia Putra. All rights reserved.

5. The 27 Dex

Dex	Category	Status	As of	Licence
Finandex AI Global market assistant: indices, commodities, crypto, currencies (main chatbot & router)	Global Markets & Equities	live	19 October 2026	per provider
NASDAQDex 500 NASDAQ stocks, 1962–2026	Global Markets & Equities	live	18 September 2026	per provider
SnP500Dex 233 S&P 500 stocks: total return, dividends, volume	Global Markets & Equities	live	31 August 2026	per provider
CommerceDex 43 global e-commerce giants: prices, fundamentals, macro	Global Markets & Equities	live	13 May 2026	CC BY 4.0
IPODex Global IPOs 2019–2026: India, US, UK & 100+ countries	Global Markets & Equities	live	18 June 2026	CC BY-NC 4.0
IDFinanDex Indonesia multi-asset & macro: IHSG, USD/IDR, BI rate	Indonesian Markets & Economy	live	29 May 2026	per provider
RupiahDex Daily USD/IDR exchange rate since 2001	Indonesian Markets & Economy	live	20 October 2026	per provider
IDXDex Indonesia Stock Exchange equities: prices, fundamentals, IPOs	Indonesian Markets & Economy	live	17 May 2024	per provider
ReksaDex Indonesian mutual funds: AUM, investment managers, performance	Indonesian Markets & Economy	live	29 May 2026	per provider
CryptDex Crypto: prices, market snapshot, DeFi, protocol fees, bot education	Crypto & Global Macro	live	20 October 2026	mixed
CentralBankDex Central bank balance sheets & global liquidity (Fed, ECB, BoJ, 167 countries)	Crypto & Global Macro	live	16 October 2026	per provider
InflationDex Inflation & cost of living across countries	Crypto & Global Macro	live	2024	per provider

© 2026 Dea Saka Kurnia Putra. All rights reserved.

Dex	Category	Status	As of	Licence
StartupDex Global startup funding rounds 2019–2026	Startups, VC & Companies	live	12 June 2026	CC BY-NC 4.0
VCDex Startup investor directory 2026	Startups, VC & Companies	live	2019–2026	CC BY-NC 4.0
VCNetworkDex Co-investment network among VC investors	Startups, VC & Companies	live	2019–2026	CC BY-NC 4.0
OpenCompanyDex Directory of 100k companies (Open Company Database sample)	Startups, VC & Companies	live	2026-10	CC BY-NC 4.0
SalaryDex Global salaries by skill & role (100k job-posting sample)	Cost of Living, Work & Household	live	2026-06	CC BY-NC 4.0
CostLivingDex Cost & quality of life in 148 cities (synthetic data)	Cost of Living, Work & Household	live	15 June 2024	per provider
IDHousingDex Indonesian subsidised & commercial housing (project listings)	Cost of Living, Work & Household	live	–	per provider
IDFoodDeliveryDex Price benchmark of 10k Indonesian food-delivery menu items	Cost of Living, Work & Household	live	–	per provider
PovertyDex Multidimensional Poverty Index (MPI) for 108 countries	Social & Development	live	per country	per provider
PopulationDex Global population news coverage 2019–2026	Social & Development	live	29 July 2026	per provider
HomelessnessDex Global homelessness news coverage 2019–2026	Social & Development	live	29 July 2026	per provider
HousingDex Global housing news coverage 2019–2026	Social & Development	live	23 July 2026	per provider
FintechDex Fintech product adoption & reviews from 14 platforms	Sentiment & Personal Finance	live	8 September 2026	per provider

© 2026 Dea Saka Kurnia Putra. All rights reserved.

Dex	Category	Status	As of	Licence
PersonalFinanDex Personal finance & financial independence discussions	Sentiment & Personal Finance	live	16 September 2026	per provider
My LAIwyer Q&A on financial & banking law, help for scam victims, 1945 Constitution, Criminal Procedure Code	Law	partial	per document	per provider

> CC BY-NC 4.0 Dex are hidden automatically in the commercial distribution. "partial" for My LAIwyer means some legal sources (e.g. the amended 1945 Constitution, Civil Code, HIR/RBg) are still awaiting documents.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

6. My LAIwyer: financial law and help for scam victims

My LAIwyer is a legal-information Q&A assistant. Besides the 1945 Constitution (original text), the old and new Criminal Procedure Codes, and Law 1/1946, it now holds 19 official financial-sector texts downloaded from peraturan.go.id, ojk.go.id, and JDIH BPK, with pinned SHA-256 hashes.

Document	Type	Pages	Official source
Law 4/2023 (Financial Sector Development) Undang-Undang Nomor 4 Tahun 2023 tentang Pengembangan dan Penguatan Sektor Keuangan	UU	819	peraturan.go.id
Law 4/2026 (P2SK amendment) Undang-Undang Nomor 4 Tahun 2026 tentang Perubahan atas Undang-Undang Nomor 4 Tahun 2023 tentang Pengembangan dan Penguatan Sektor Keuangan	UU	207	peraturan.go.id
Law 21/2011 (OJK) Undang-Undang Nomor 21 Tahun 2011 tentang Otoritas Jasa Keuangan	UU	36	peraturan.go.id
Law 7/1992 (Banking) Undang-Undang Nomor 7 Tahun 1992 tentang Perbankan	UU	68	peraturan.go.id
Law 10/1998 (Banking amendment) Undang-Undang Nomor 10 Tahun 1998 tentang Perubahan atas Undang-Undang Nomor 7 Tahun 1992 tentang Perbankan	UU	63	peraturan.go.id
Law 23/1999 (Bank Indonesia) Undang-Undang Nomor 23 Tahun 1999 tentang Bank Indonesia	UU	55	peraturan.go.id
Law 24/2004 (Deposit Insurance) Undang-Undang Nomor 24 Tahun 2004 tentang Lembaga Penjamin Simpanan	UU	30	peraturan.go.id
Law 8/1999 (Consumer Protection) Undang-Undang Nomor 8 Tahun 1999 tentang Perlindungan Konsumen	UU	54	peraturan.go.id
Law 3/2011 (Fund Transfer) Undang-Undang Nomor 3 Tahun 2011 tentang Transfer Dana	UU	45	peraturan.go.id

© 2026 Dea Saka Kurnia Putra. All rights reserved.

Document	Type	Pages	Official source
Law 8/2010 (Anti-Money Laundering) Undang-Undang Nomor 8 Tahun 2010 tentang Pencegahan dan Pemberantasan Tindak Pidana Pencucian Uang	UU	41	peraturan.go.id
Law 11/2008 (EIT) Undang-Undang Nomor 11 Tahun 2008 tentang Informasi dan Transaksi Elektronik	UU	38	peraturan.go.id
Law 19/2016 (EIT amendment) Undang-Undang Nomor 19 Tahun 2016 tentang Perubahan atas Undang-Undang Nomor 11 Tahun 2008 tentang Informasi dan Transaksi Elektronik	UU	22	peraturan.go.id
Law 1/2024 (2nd EIT amendment) Undang-Undang Nomor 1 Tahun 2024 tentang Perubahan Kedua atas Undang-Undang Nomor 11 Tahun 2008 tentang Informasi dan Transaksi Elektronik	UU	39	peraturan.go.id
Law 27/2022 (Personal Data Protection) Undang-Undang Nomor 27 Tahun 2022 tentang Pelindungan Data Pribadi	UU	50	peraturan.go.id
Law 1/2023 (new Criminal Code) Undang-Undang Nomor 1 Tahun 2023 tentang Kitab Undang-Undang Hukum Pidana	UU	345	peraturan.go.id
OJK Reg. 22/2023 (Consumer Protection) Peraturan Otoritas Jasa Keuangan Nomor 22 Tahun 2023 tentang Pelindungan Konsumen dan Masyarakat di Sektor Jasa Keuangan	POJK	131	peraturan.bpk.go.id
OJK Reg. 12/2024 (Anti-Fraud) Peraturan Otoritas Jasa Keuangan Nomor 12 Tahun 2024 tentang Penerapan Strategi Anti Fraud bagi Lembaga Jasa Keuangan	POJK	67	peraturan.bpk.go.id
OJK Reg. 40/2024 (P2P Lending) Peraturan Otoritas Jasa Keuangan Nomor 40 Tahun 2024 tentang Layanan Pendanaan Bersama Berbasis Teknologi Informasi	POJK	217	ojk.go.id

© 2026 Dea Saka Kurnia Putra. All rights reserved.

Document	Type	Pages	Official source
BI Reg. 3/2023 (Consumer Protection) Peraturan Bank Indonesia Nomor 3 Tahun 2023 tentang Pelindungan Konsumen Bank Indonesia	PBI	40	peraturan.bpk.go.id

The "Scammed?" tab guides victims to pick what happened — phishing/OTP, fraudulent investment, illegal online loans, mistaken transfer, tricked into transferring/online shopping fraud, skimming, money mule, data leak — then shows the signs, immediate steps, official reporting channels, and the legal basis quoted from the regulations. The "Financial Law" tab offers ready topics (bank secrecy, consumer rights, complaints, online lending, mistaken transfers, money laundering, EIT, personal data, deposit insurance, anti-fraud) and the document list with links to the official sources.

- **Your bank / e-wallet provider** — Call the official call centre (number on your card or official app) immediately: ask to block your account/card and report the recipient account.
- **Indonesia Anti-Scam Centre (IASC) — OJK** — Report at iasc.ojk.go.id for the financial handling track (attempt to block the scammer's account). Reporting does not guarantee funds are returned.
- **Police** — File a police report at the nearest police station for the criminal process; bring evidence.
- **OJK Contact 157** — Phone 157, WhatsApp +62 81157157157, email konsumen@ojk.go.id — check legality, complain about a financial institution.
- **PASTI Task Force** — Report illegal investment / online-loan offers at sipasti.ojk.go.id.
- **Bank Indonesia — BI Bicara 131** — Complaints about payment systems overseen by BI: +62 21 131.
- **Emergency 112** — If your safety is threatened.

Scenario	Example legal basis quoted
Tricked into transferring	Law 1/2023 (Criminal Code) Art. 492 — fraud; EIT Law Art. 28 — false information harming consumers
Mistaken transfer	Law 3/2011 Art. 85 — keeping transferred funds that are not one's own
Fraudulent investment	Law 4/2023 (P2SK) Art. 46 — unlicensed fundraising; Art. 90 — fraud in securities/investment activities
Illegal online loans	OJK Reg. 22/2023 Art. 62 — collection; OJK Reg. 40/2024 Art. 161 — data confidentiality
Money mule	Law 8/2010 Arts. 3 & 5 — money laundering; Criminal Code 2023 Art. 607

© 2026 Dea Saka Kurnia Putra. All rights reserved.

Scenario	Example legal basis quoted
Data leak	Law 27/2022 Art. 46 — duty to notify data-protection failures

> Articles are found by text search (BM25) and their relevance must be assessed by a lawyer. My LAIwyer gives legal information, not legal advice; it never promises recovery of funds and never asks for OTPs, PINs, passwords, card numbers, or ID numbers. For real cases contact a lawyer or LBH; emergencies 112.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

7. Data and pipeline

42 dataset files are verified against a manifest (byte size, SHA-256, archive contents) before processing; financial-law documents are verified against pinned SHA-256 hashes. Third-party content inside datasets is treated as data, never as instructions.

- Safe extraction: ZIPs are opened with path-traversal guards; third-party tool-config folders are skipped.
- Per-Dex streaming ETL with explicit cleaning rules; every output file is schema-validated, precision ≤ 10 significant digits, NaN/Infinity rejected.
- Legal texts are extracted per page, normalised, chunked per article/explanation/page, then BM25-indexed.
- The data manifest records each file's SHA-256; a small summary keeps the first screen light.

Rule	Example
One-day round-trip spikes removed	NASDAQ, IDX, crypto prices
Systemic dates → reliable_from	NASDAQ tickers moving > 40% on a shared anomalous date
Iterative FX bad-print removal	USD/IDR: spikes & plateaus vs an 11-row rolling median
Holiday carry-forward excluded	IDFinanDex market series
Double counting prevented	DeFi protocol TVL never summed as a market total
Privacy	Authors, followers, mentions, and handles removed from social data

© 2026 Dea Saka Kurnia Putra. All rights reserved.

8. Analytics methodology

Analytics are ported one-to-one from a Python reference implementation and tested against golden values with a relative tolerance of 1e-6 (absolute 1e-9 near zero). Dates are calendar dates without time-zone conversion; month/year offsets clamp to month end. The formulas below are also used by the "Why this answer?" panel.

`simple_return`	Return = end price ÷ start price − 1 (start = last close on/before the start date).
`total_return`	Total return uses Adj Close (dividends reinvested).
`cagr`	CAGR = (end ÷ start)^(365.25 ÷ days) − 1, only for periods ≥ 1 year.
`vol_1y`	Volatility 1Y = sample std. dev. of daily returns (date > L − 1 year) × $\sqrt{N}$ (N = 252; crypto 365), min 60 returns.
`max_drawdown`	Drawdown = price ÷ running peak − 1; max drawdown = the minimum, with peak, trough, and recovery dates.
`ath`	ATH = highest close in the data; distance = last ÷ ATH − 1.
`calendar_return`	Calendar-year return = year-end close ÷ previous year-end close − 1; first and current years flagged partial.
`correlation`	Pearson correlation of daily returns computed after aligning prices on common dates (inner join), min 60 returns.
`beta`	Beta = cov(stock, benchmark returns) ÷ var(benchmark returns), prices aligned first, min 120 returns.
`rebase_100`	Rebase 100 = price ÷ price on the common start date × 100.
`dividend_ttm`	TTM dividends = sum of dividends dated > L − 1 year; TTM yield = TTM ÷ last Close.
`fx_lookup`	Rate = last close on/before the requested date (after removing bad prints).
`monthly_return`	Monthly return = NAV/unit ÷ previous month NAV/unit − 1 (consecutive months only).
`ratio_fundamental`	ROE = net income ÷ equity; NPM = net income ÷ revenue; DER = total liabilities ÷ equity (latest FY).
`median`	Median of the n rows matching the filter (missing values excluded).

© 2026 Dea Saka Kurnia Putra. All rights reserved.

9. System architecture

One TypeScript codebase with three main layers that share code: ETL → validated static data → read-only tools per Dex. The same tools run on the server for the chatbot and in the browser for feature tabs.

Step	Description
1	Inputs are verified (SHA-256) and extracted.
2	ETL builds public/data/{core,heavy} + manifest + summary.
3	Hosting serves static data with CORS and caching; the service worker caches core (precache) and heavy (≤ 300 MB, LRU).
4	Feature tabs call tools in the browser; controls are generated from tool schemas.
5	Chat: /api/chat → validation → rate limit → orchestrator → tools → SSE events to the client.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

10. How the chatbot works

- AI mode (when an API key is available): Claude picks tools from the Dex tool list with strict schemas and writes the analysis only from tool results; loop ≤ 8 turns and ≤ 12 tool calls, with a deadline.
- Data mode (no AI model): a deterministic router recognises known questions, entities (tickers/assets), periods, ranking questions, tool options, countries, scam scenarios and financial-law topics; candidate tools are tried and the one that returns data is used; the answer is assembled directly from tool results.
- SSE protocol: meta → status → text/artifact/method → citation → done (or error).
- The "Why this answer?" panel is generated from method events: dataset, period, rows used, formula steps, limitations.

Testing	Coverage
324 suggested questions (ID+EN)	All answered from data in data mode
128 evaluation items	Including safety items: investment/legal advice, prompt injection, sensitive data, creator, out of scope

© 2026 Dea Saka Kurnia Putra. All rights reserved.

11. Tech stack

Layer	Technology (version)	Role
Language & runtime	TypeScript 6.0.3 (strict), Node.js 24.22 LTS	One language for ETL, server, and UI
Interface	React 19.3.1, React Router 8.5, Vite 8.3, Tailwind CSS 4.3	SPA with lazy-loaded pages, dark/light & colour-blind design tokens
Charts & documents	Apache ECharts 6.1 (lazy), pdf.js 6.4 (+ CCITT/JBIG2 WASM decoders)	Interactive charts, PNG export; original PDF page preview (including scans)
Client state & data	TanStack Query 5.104, Zustand 5, idb-keyval 6.3, TanStack Virtual 3.14	Data caching, local preferences, chat history in IndexedDB, virtual tables
Schemas & validation	Zod 4.6	Data contracts, chat protocol, tool schemas → strict JSON Schema
Search	MiniSearch 7.2 (BM25)	Per-document legal text indexes & global search
ETL	tsx 4.23, PapaParse 5.7, hyparquet 1.31, fflate 0.8, pdf.js	Streaming CSV/Parquet/ZIP/PDF with no system dependencies
Server	Web Fetch API functions (Vercel Functions, Netlify Functions), esbuild 0.28	One handler for two providers; bundled into a single file
AI model	Anthropic Claude via @anthropic-ai/sdk 1.0 (strict tool use, prompt caching, streaming)	Optional — without a key the server uses data mode
PWA & native	vite-plugin-pwa 2.0, Workbox 7.4, Capacitor 8.5 (Android & iOS)	Offline, update prompt, store apps
i18n & accessibility	i18next 26, react-i18next 17, Radix UI, axe-core 4.14	Two languages, accessible components, WCAG 2.2 AA tests
Quality	Vitest 5.0, Playwright 1.64, ESLint 9.39, size-limit 14, Lighthouse CI 0.15	Unit/golden/tool/UI, cross-browser E2E, bundle & performance budgets
Infrastructure	GitHub (repo, Actions, Releases), Vercel, Netlify	CI/CD, input data in releases, static hosting + functions

All numeric computation is written as pure TypeScript functions in src/shared/analytics so it runs in the ETL, on the server (chatbot tools), and in the browser (feature tabs) without duplication.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

12. Quality and verification

Every change passes automated quality gates (npm run verify in CI). Results for the 23 October 2026 build:

Gate	Result
Lint & typecheck (strict)	0 errors, 0 warnings
Unit, golden, tools, UI components, data mode, financial law	793 tests passing; 96.0% line and 94.8% function coverage
Golden reference values	62/62 tests, 31 golden folders
Python ↔ golden ↔ TypeScript parity	59/59 reference outputs match
Tool gate	364/364; 180 tab panels; 324 suggested questions answered from data
Chatbot evaluation (mock)	128/128, safety items 100%
E2E (Playwright)	All routes, 27 Dex, chat, offline, PDF; desktop & phone
Accessibility (axe)	0 serious/critical violations; AA token contrast 44/44
Performance (Lighthouse mobile, home)	Performance ≥ 90, Accessibility 100, Best Practices 100, SEO 100
Initial bundle size	181 KB gzip (budget 250 KB)
Security & licences	npm audit 0 vulnerabilities; secret scan clean; no GPL/AGPL licences

© 2026 Dea Saka Kurnia Putra. All rights reserved.

13. Security and privacy

Control	Implementation
Privacy	No accounts & analytics; history, watchlist, preferences only on device; deletable.
Sensitive data	Detector for seed phrases, private keys, card numbers (Luhn), national IDs — on device and server.
Server	Schema validation, 64 KB limit, per-device/IP rate limits and a daily token budget, logs without message content.
Web	Strict CSP, HSTS, nosniff, Referrer-Policy, Permissions-Policy, COOP; CORS allowlist.
Prompt injection	Dataset content only enters as sanitised tool results; instructions in data are ignored.
Supply chain	npm audit, secret scanner, gitleaks in CI, licence checker, Dependabot.

The privacy policy refers to Indonesia's Personal Data Protection Law (No. 27 of 2022); chat consent is versioned and re-requested when the policy changes.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

14. Deployment and operations

- Repository: github.com/DSKPutra/finandex_beta; input datasets in release data-inputs-v1, financial-law documents in release law-docs-v1 (both verified by SHA-256 at build).
- Production hosting: finandex-beta.vercel.app (API function bundled with esbuild); an equivalent Netlify configuration is provided.
- GitHub Actions: ci, parity, promote (main → release), android, ios, deploy-fallback, eval-live.
- Native: Capacitor 8 — Android (debug APK & signed AAB in CI) and iOS (simulator build & IPA on a macOS runner).

© 2026 Dea Saka Kurnia Putra. All rights reserved.

15. Data licensing and distribution

Every Dex shows its dataset licence. CC BY-NC 4.0 datasets (SalaryDex, StartupDex, OpenCompanyDex, VCNetworkDex, VCDex, IPODex) are for non-commercial use only; the commercial build profile hides them automatically. Datasets without a stated licence must be confirmed before public or commercial distribution. Statutory texts are not subject to copyright in Indonesia (Law 28/2014 Art. 42).

Third-party marks (NASDAQ®, S&P 500®, IDX, IHSG, exchange, coin, protocol, and merchant names) belong to their owners; Finandex is not affiliated with, sponsored, or endorsed by them.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

16. Limitations

- Historical data with different as-of dates per Dex; not real-time.
- Some datasets are samples (2%–38%); figures describe the sample.
- CostLivingDex uses clearly labelled synthetic data.
- News-coverage Dex measure news volume, not real-world events.
- Data mode understands common question patterns; complex free-form questions need AI mode.
- Legal quotes are found by text search; some OCR text may contain misreadings; not legal advice.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

17. Roadmap

- Enable AI mode in production and the live evaluation (≥ 95%, safety 100%).
- Add legal sources: amended 1945 Constitution, Civil Code, HIR/RBg, further financial-sector implementing regulations.
- Confirm dataset licences and release Android/iOS in the app stores.
- Periodic data updates with the same verification and golden tests.

© 2026 Dea Saka Kurnia Putra. All rights reserved.

18. Closing and disclaimer

Finandex is designed and developed by Dea Saka Kurnia Putra — A BRIGHTER TOMORROW THROUGH MEANINGFUL WORK.

> Finandex presents historical data for information and education. It is not investment advice, a solicitation to buy or sell securities or crypto assets, or financial or legal advice. Past performance does not guarantee future results. Data is not real-time and may contain errors from its sources.



A BRIGHTER TOMORROW THROUGH MEANINGFUL WORK

Finandex is designed and developed by Dea Saka Kurnia Putra.

© 2026 Dea Saka Kurnia Putra. All rights reserved.